

A. General Information

Information Requested	Information provided by the PI
1. Heritage Craft Name	Negamam Cotton Saree
2. Affiliated Institute	Kumaraguru College of Technology, Coimbatore
3. Principal Investigators	Dr. V. Ramesh Babu
4. Official Project Title	Reviving Heritage Textiles through Sustainable Innovation: Negamam Sarees
5. Clusters Visited	Vadambacherry, Negamam, Peria Negamam, Samathur, Arani, Varanasi

1. Introduction

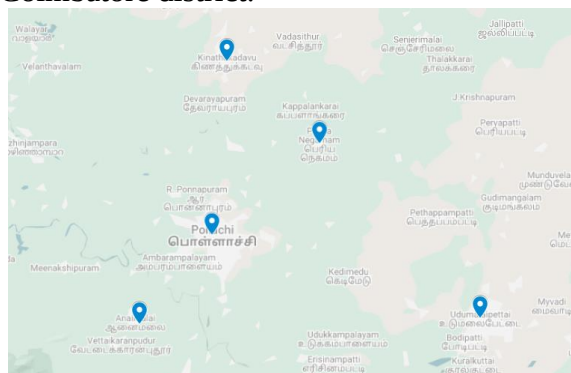
• Brief description of the textile product

Negamam Cotton Sarees are handloom sarees produced in the Coimbatore region of Tamil Nadu, where the moisture-laden winds from the Western Ghats support the weaving process. The sarees are typically woven using 80s combed cotton, which gives the fabric a dense, durable, and comfortable feel suitable for different climates. Production is usually concentrated between June and November, when the humidity is favourable for handling the yarn.

A key feature of these sarees is the richly woven pallu, often designed with traditional motifs such as peacocks, parrots, paisleys, elephants, and swans, arranged in repeated or mirrored patterns. Jacquard techniques are commonly used to achieve these detailed designs, and in some cases, half-fine zari is incorporated to add subtle richness to the overall appearance.

• Geographical area of production (map if possible)

Production of Negamam handloom sarees is centered in Negamam village, Pollachi Taluk, Coimbatore district, Tamil Nadu (around 10.7333°N, 77.1°E), with the cluster spread across the Pollachi - Negamam belt and extending to nearby villages. It is manufactured by 15 weavers' societies across villages of Pollachi, Sulur, Kinathukadavu, Anamalai and Udumalpet Taluks in Coimbatore district.



- **Key characteristics (material, technique, motifs, end-use)**

- **Material:** 80s combed cotton (warp/weft); Zari or mercerized cotton for extra-weft designs.
- **Technique:** Exclusively handloom (pit/raised-pit looms) using traditional "Thandi" attachments for high density, natural starches for sizing, and vat dyeing.
- **Motifs:** Traditional icons like the peacock (Annam), Rudraksha, and various Temple (Komai) borders.
- **End-use:** Historically an 8-meter drape for Kongu women; currently a 6.2-meter standard saree for daily and functional wear.

- **Why is it classified as a *textile heritage***

Negamam Cotton Sarees are classified as a textile heritage due to their long-standing association with the traditional handloom practices of the Devanga weaving community, whose weaving lineage in South India dates back several centuries. The craft in the Negamam–Pollachi region itself has been sustained for over 200 years, with weaving practiced continuously across generations within families.

The sarees are produced using time-tested techniques such as manual warping, sizing, and handloom weaving, which have been preserved with minimal technological intervention. Knowledge and skills are traditionally transmitted within households, often spanning two to three generations or more, reinforcing its continuity as a living heritage practice.

The craft also reflects a deep connection to the local environment, as climatic conditions in the Pollachi region support cotton weaving processes, along with strong ties to cultural practices and everyday life. Characterised by simple designs, durable textures, and nature-inspired motifs, Negamam sarees embody both material and intangible heritage, representing the regional identity and socio-cultural fabric of the community.

- **Cultural and traditional significance**

Negamam Cotton Sarees hold strong cultural significance within the Devanga weaving community, where weaving is not only a livelihood but also a tradition passed down through generations. The craft is closely connected to community practices, rituals and a shared sense of identity among the weavers.

According to a traditional legend, Lord Brahma first created Manu and instructed him to weave garments for both the Devas and humans. After fulfilling this duty, Manu is said to have attained a higher spiritual state. In his absence, the Devas were left without proper clothing and are believed to have suffered, covering themselves with leaves. They then prayed to Brahma, who took them to Lord Siva. In response, Lord Siva created Devalan, a divine weaver, to clothe the Devas. It is believed that the descendants of Devalan came to be known as the Devangas, meaning “those who are part of the divine body”.

This legend reinforces the idea of weaving as a sacred and respected duty within the community. Even today, this belief is reflected in their practices, where work is often aligned

with ritual observances and respect for their community deity. The sarees themselves are an important part of everyday life in the Kongu region, valued for their durability and practicality.

- **Current market presence (local/national/export)**

While the sarees are primarily sold in domestic and national markets, a portion reaches international markets. They are known to reach countries such as the USA, UK, Canada, Australia, Singapore, Malaysia, and parts of the Middle East.

It is estimated that around 2.25 lakh cotton sarees are produced annually through handloom weaver's cooperative societies, with a production value of approximately ₹10 crores. The annual sales turnover is about ₹12 crores.

2. History

- **Origin of the craft (region/community)**

Negamam Cotton Sarees originates from Negamam village in Pollachi Taluk of Coimbatore district. The village began to develop after 1871, when it was granted a permanent sanad and gradually became a centre for weaving and trade. The craft is closely linked to the Devanga (Devanga Chettiar) community, a traditional group of skilled weavers known for producing cotton and silk textiles. They migrated to this region from different parts of the Kongu belt and continued their weaving practices here. Over time, the sarees produced in Negamam gained popularity in the surrounding areas and the product came to be identified by the name of the village itself.

- **Historical development**

In its early phase, Negamam functioned as a local weaving and trading hub, supplying cotton sarees to nearby villages. Traditionally, weaving communities in the Kongu region produced coarse coloured sarees using lower-count cotton yarns such as 20s and 30s. Gradually, there was a shift toward finer counts like 40s, 60s, and 80s, improving the quality and durability of the fabric. The Devanga weavers played a significant role in this development and their sarees became well known for their strength and usability.

As demand increased, weaving activities spread to nearby villages, expanding the reach of Negamam sarees across the Coimbatore region. In the mid-20th century, the formation of cooperative societies helped organize production and support the weavers. Societies established from 1942 onwards played an important role in providing yarn, ensuring consistent production and facilitating market access. Through these developments, Negamam Cotton Sarees maintained their identity while adapting to changing needs over time.

- **Major transformations over time (materials, techniques, designs)**

- **Dimensional Shift:** The traditional 8-meter length has largely transitioned to the modern 6.2-meter standard (including blouse) as draping styles modernized.
- **Technical Density:** While historical variants maintained a density of 86–90 PPI, contemporary market-grade versions often range between 70–76 PPI, though the use of 80s combed cotton remains the technical benchmark.

- **Design Evolution:** Simple dobby-based geometric patterns and narrow borders have expanded into complex Jacquard-driven layouts featuring elaborate extra-weft motifs and high-contrast color palettes.
- **Timeline (mandatory): Ancient → Colonial → Post-independence → Present**

Ancient

The weaving tradition in this region is linked to the Devanga community, known for their skills in cotton and silk weaving. Early production in the Kongu region mainly involved coarse cotton sarees using lower yarn counts, which gradually improved over time. Negamam functioned as a local centre for weaving and trade.

Colonial (Late 19th Century)

After the 1871 Permanent Sanad, Negamam became a settled weaving cluster. The Devanga weavers migrated here and established production, supported by favourable climatic conditions, especially the humidity from the Western Ghats that aided fine-count cotton weaving.

Post-independence (1900s – 2010s)

This period marked both growth and change. The durable 8-meter saree became widely used in the Kongu region. With the formation of cooperative societies from the 1940s, production became more organized. From the 1980s onwards, the rise of powerlooms led to increased competition, resulting in a shift to the 6.2-meter saree and gradual design changes.

Present

Currently, production continues with a reduced number of weavers compared to earlier periods. The focus remains on maintaining quality while adapting to changing market preferences. However, challenges such as rising costs and inconsistent demand persist, alongside some support from cooperatives and growing interest in handloom products.

- **Influence of government policies, industrialization, or globalization**

Government support through cooperative societies has helped sustain the handloom sector to some extent. At the same time, industrialization especially the growth of powerlooms has created strong competition by offering cheaper and faster alternatives.

3. Manpower

- **Number of families/artisans currently involved**

The weaving activity is spread across multiple villages and involves various pre-loom processes carried out by different groups. The cluster has a substantial number of looms (estimated to be above 4,500 approx), reflecting strong involvement of households both directly and indirectly in the craft.

- **Comparison with past (10–20 years ago)**

Over the last 10–20 years, there has been a visible contraction in the workforce. At its peak in the late 20th century, the cluster was the primary economic driver for the region. Today, the

number of active looms has declined as the surrounding area (Pollachi/Coimbatore) has industrialized, offering alternative employment.

- **Gender-wise and age-wise participation**

- **Age-wise:** Participation is heavily skewed toward an aging workforce. Weavers often fall in the 40 to 80 years age bracket.
- **Gender-wise:** The craft is a family-integrated effort. Men typically manage the high-tension loom operations and jacquard setups, while women handle critical pre-loom processes including pirn winding and warping in addition to weaving.
- **Generational Gap:** There is minimal participation from the younger generation (<35 years), who are largely migrating toward urban professional roles.

- **Skill levels (master artisans, helpers, trainees)**

Involves various skilled roles such as master weavers, dyers, designers and specialized artisans. While older artisans possess high levels of expertise developed over generations, the newer generation often lacks comparable skill depth. This creates a need for stronger skill development, knowledge transfer and greater exploration of new techniques and design approaches.

- **Average income per household (range)**

Around Rs. 20000 to 25000 approximately

- **Livelihood dependency (% income from this craft)**

Almost all active weaving families are primarily dependent on handloom weaving and related processes for their livelihood, with nearly 90–100% of their household income coming from this craft. Other sources of income are minimal or occasional.

- **Reasons for increase/decrease in workforce**

Decrease in workforce due to -

- Long daily working hours with sustained physical effort.
- High manual labour input across warping, sizing, and loom operation.
- Low wage realisation relative to skill and labour invested.
- Income instability due to market fluctuations and powerloom competition.

- **Migration trends (if any)**

There is a gradual migration of younger members toward nearby urban centers for employment and education. Both seasonal and permanent migration are observed, driven by the need for more stable and higher-paying jobs

- **Key challenges faced by artisans**

- Rising cost of raw materials like yarn reduces profit margins and overall production
- Family members are required for pre-processing work but are less paid or acknowledged

- Continuous physical strain leads to health issues especially among elderly artisans
 - The daily regime is grueling, often starting at 5 a.m. and involving sustained physical torque that becomes difficult for the aging artisan population to maintain.
 - Knowledge is still passed down informally, but the younger generation is increasingly opting for more stable, less physically demanding jobs.
 - Earnings are often low compared to the effort and time involved
- **Expectations/aspirations of the community**
 - Want better and more stable income for the work they put in
 - Hope for reduced physical strain with some support or improved tools
 - Expect more recognition and respect for their craft
 - Need easier access to loans, subsidies, and financial support
 - Look for steady demand and regular work throughout the year

4. GI tag

- **Why GI tag given**

The GI tag was given to protect the identity of Negamam Cotton Sarees. These sarees are known for their dense weave, use of fine 80s combed cotton and dependence on the unique geo-climatic conditions of the Negamam region that supports the process. The tag also helps prevent copies made on powerlooms.

- **Unique features**

The unique identity of Negamam Cotton Sarees is defined by the combined influence of raw material selection, traditional handloom practices, geo-climatic conditions and continued relevance in modern times. These elements together form the core identifying techniques that distinguish the saree from other cotton handloom products.

1. Raw Material–Based Technique

Negamam Cotton Sarees are produced using cotton yarn supplied through NHDC cooperative societies, selected for its fineness and suitability for handloom weaving. The yarn enables high breathability and comfort, making the fabric well suited to Tamil Nadu's hot and humid climate. This results in a saree that is lightweight, airy and comfortable for daily wear.

Core Technique:

Use of fine-count cooperative cotton yarn optimized for breathable handloom fabric.

2. Traditional Handloom Technique

Unlike most modern cotton saree production, Negamam Cotton Sarees are woven exclusively on handlooms, without the use of power looms. The entire production process - spinning, hank dyeing, warping, natural sizing and weaving is carried out using traditional methods. These skills are preserved through intergenerational knowledge transfer and only a limited number of skilled weavers in the Coimbatore region continue this practice.

Core Technique:

Exclusive handloom production using inherited skills and non-mechanized processes.

3. Natural Yarn Treatment Technique

The warp yarn is treated using natural sizing materials such as rice gruel, maize and potato starch. This traditional sizing method improves yarn strength while maintaining softness and flexibility, contributing to the durability and hand feel of the fabric.

Core Technique:

Application of natural sizing methods for balanced strength and softness

4. Geo-Climatic Technique

The location of Negamam near the Western Ghats exposes the weaving process to seasonal monsoon humidity. The moisture-laden winds during June to September create favorable conditions for yarn handling and weaving, contributing to the saree's soft yet durable texture. Production is therefore concentrated between June and November.

Core Technique:

Seasonal production aligned with monsoon-driven climatic conditions.

5. Technique of Contemporary Adaptation

While retaining its traditional weave structure and production methods, the Negamam Cotton Saree has adapted to present-day preferences through subtle design and colour variations. This balance between tradition and adaptability has increased its acceptance among modern consumers without altering its core identity.

Core Technique:

Design adaptability without change to traditional weave structure.

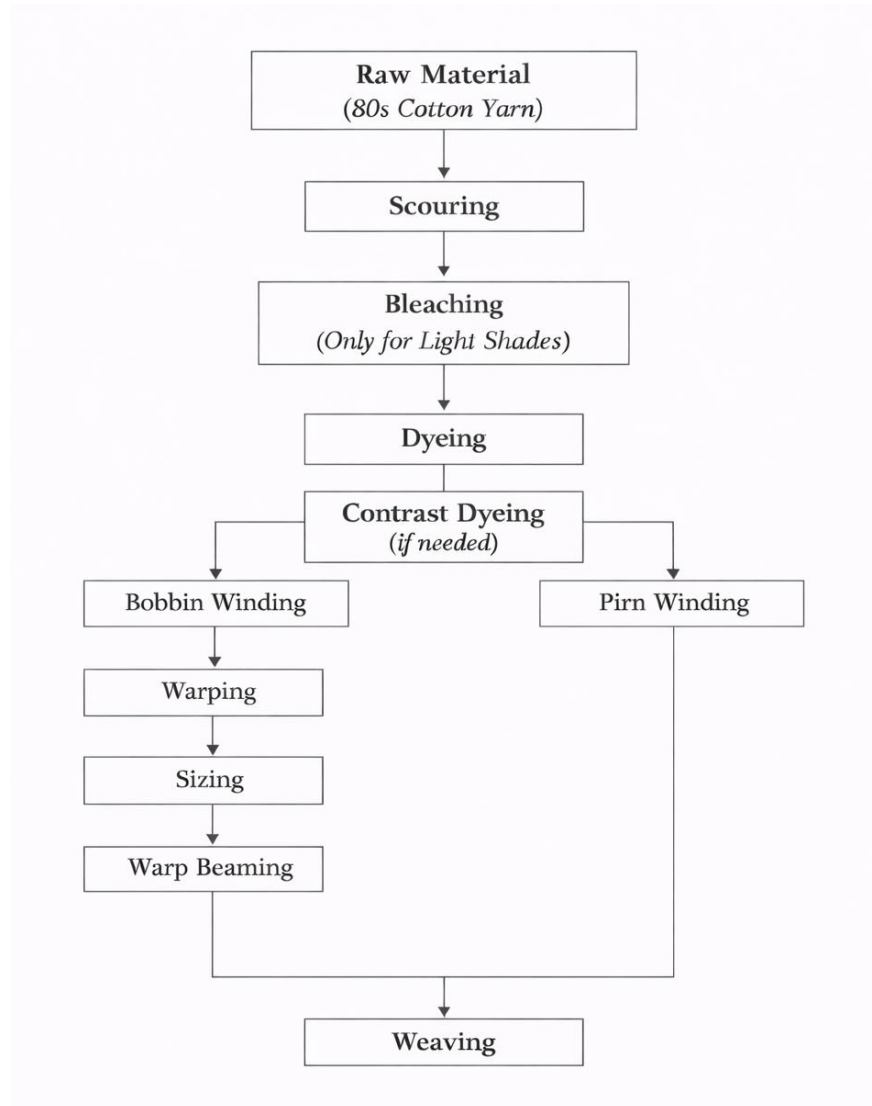
Dense plain weave using 80s cotton with 86–90 picks per inch, extra-weft motifs woven on dobby/jacquard, and use of raised-pit looms with Thandi to achieve higher fabric density.

- **Specify the GI Tag document no.**

Geographical Indication Number: 766 (year: 2022 – 2023)

5. Manufacturing steps

- **Detailed manufacturing process with a flowchart and images.**



Negamam Cotton saree production process

Step 1: Selection and Procurement of Yarn:

The production of Negamam cotton sarees begins with the selection of 80s count cotton yarn, which is known for its fineness and suitability for handloom weaving. The yarn is sourced either from the National Handloom Development Corporation (NHDC) through cooperative societies or directly from spinning mills and yarn dealers by private weavers. As per handloom norms, yarn supplied to cooperative societies is dyed only in government-approved dyeing units to maintain quality and consistency.

Step 2: Boiling Yarn:

Once the yarn is received, the bundles are separated and subjected to boiling. This process is carried out using a solution containing TT oil, caustic soda and soda ash. Boiling helps in removing natural impurities, oils and dirt present in the cotton yarn, making it clean and suitable for further processing.

Step 3: Washing and Manual Squeezing:

After boiling, the yarn is washed only in clean, plain water to remove loosened impurities. No chemicals are used in this washing stage. The washed yarn is then manually squeezed to remove excess water, ensuring the yarn remains soft and ready for dyeing.

Step 4: Dyeing:

Yarn Dyeing is carried out using vat dyes to achieve the required colour shades. For lighter shades, the yarn undergoes bleaching before dyeing, while darker shades are dyed directly after scouring. A dye bath is prepared using hot water, caustic soda, hydrosulphate and selected dyes. The yarn is measured and tied into bundles (paavu) and immersed in the dye bath for approximately 30 minutes. The yarn is turned at regular intervals to ensure even colour penetration. After dyeing, the yarn is washed in clean water and dried in the shade for several days. Shade drying allows proper oxidation, resulting in stronger and long-lasting colours. The dyed yarn is then bundled for further processing.



Yarn pre-process and dyeing

Step 5: Weft Preparation through Pirn Winding:

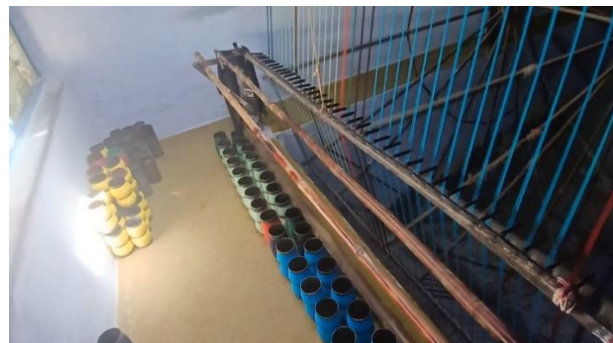
The yarn required for weft insertion is wound onto small cylindrical holders known as pirns using a hand-driven charkha. These pirns are later placed inside the shuttle during weaving.



Pirn Winding

Step 6: Warping:

Warping involves arranging the warp yarns in a parallel manner to form a warp sheet of the required length and width. The yarn is wound from bobbins and aligned systematically. The prepared warp is then taken for sizing.



Warp winding

Step 7: Sizing of Warp Yarn:

Sizing is carried out to improve the strength, smoothness and weaving performance of the warp yarn. The sizing mixture is traditionally prepared using potato starch, coconut oil, maize and

rice gruel. The sizing solution is applied evenly to the warp using a sprayer and spread with a brush. This process reduces yarn breakage during weaving. After sizing, the threads are carefully separated and aligned, usually with the help of multiple workers.

Step 8: Preparation of Warp Beam:

The sized warp is stretched in an open area such as a street or courtyard. Broken or damaged threads are repaired and the required saree length is marked. The warp is then rolled onto a wooden warp beam and transported to the weaver's loom

Step 9: Drawing-in and Denting:

In the drawing-in process, warp threads are passed through the healds of the loom in a predetermined sequence. This is followed by denting, where the threads are drawn through the reed according to the fabric structure. New warp threads are joined with the old ones using a traditional twisting method, usually done by the weaver or family members.

Step 10: Weaving:

Negamam sarees are woven using pit looms or raised pit looms fitted with fly shuttles. The looms are simple in structure and the healds are commonly made from nylon wire. The weft yarn is inserted manually by passing the shuttle across the shed, followed by beating with a suspended sley. This method ensures tight weaving and high fabric quality.

Average time taken for weaving a saree: ~ 1.5 days to 2 days



Pit Loom and Frame loom

Step 11: Quality Inspection and Packing:

After weaving, each saree is thoroughly inspected by the society appraiser. The inspection includes checking length, width, weave uniformity and defects in warp or weft. Approved sarees are folded neatly and packed for distribution.

- **How much raw materials are used per saree**

Approximately around 400 grams of cotton yarn are required per saree, depending on the density, design complexity and border work.

- **Where the raw materials are sourced**

Co-operative societies from NHDC and Private weavers from yarn dealers or spinning mills according to their requirements.

- **How much energy is utilised in each process**

Energy consumption is minimal and largely manual across stages like warping, sizing, and weaving.

- **Form of energy- wood/coal/solar**

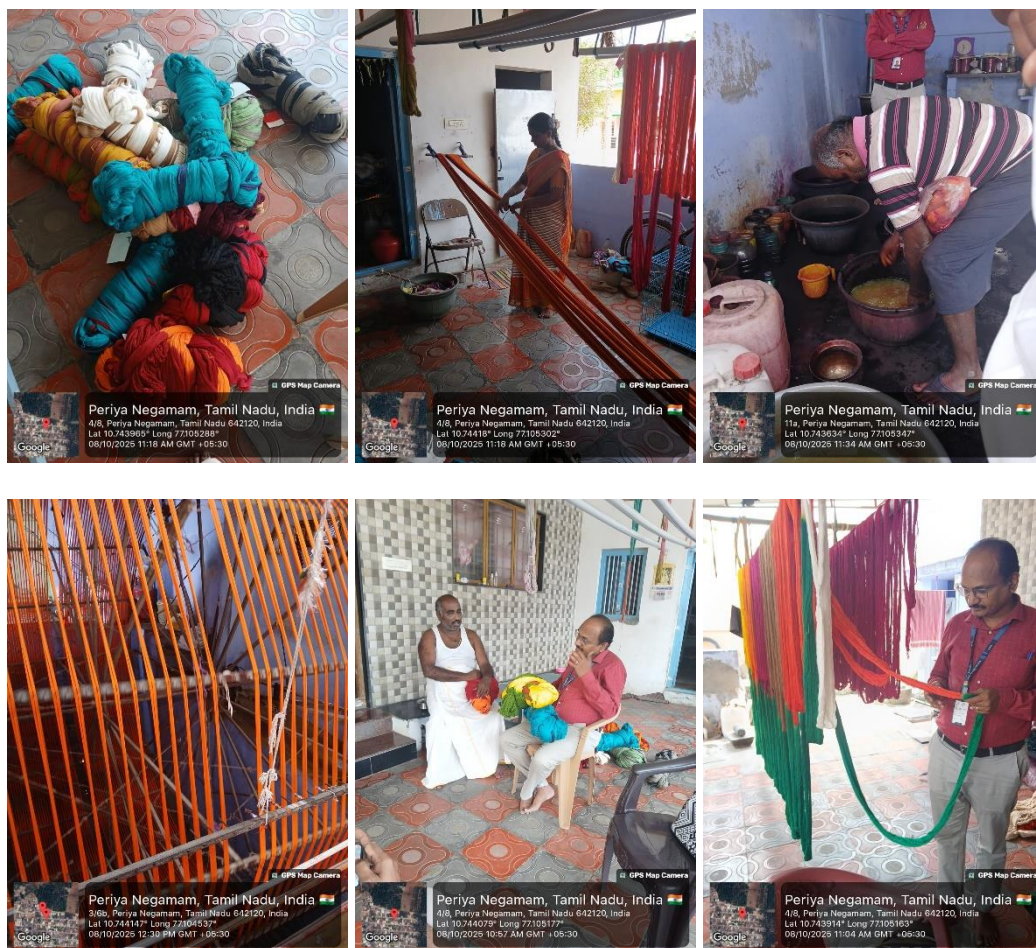
- Firewood or gas is commonly used for boiling and dyeing
- Use of electricity

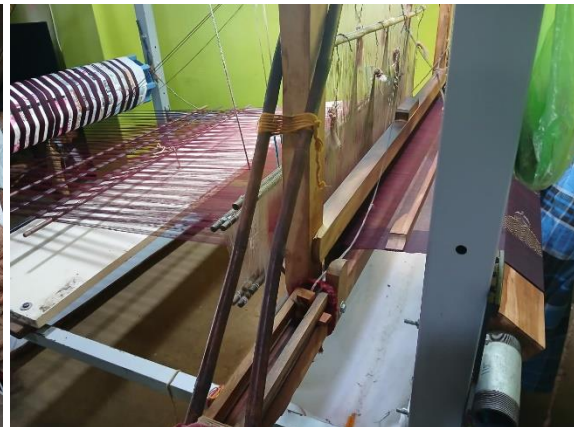
6. Real photos of Cluster /Product/manufacturing process (Geo tagged if possible)

Field visit Photos:

Negamam field visit:

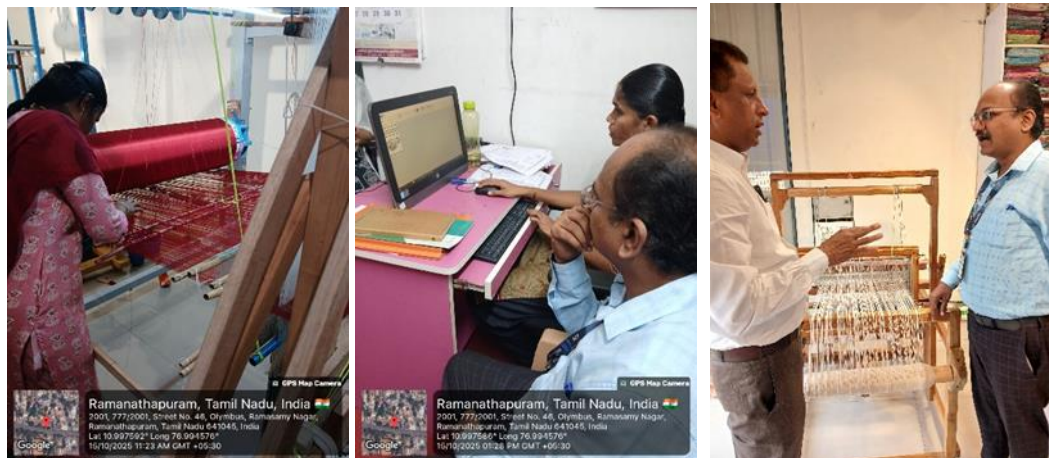
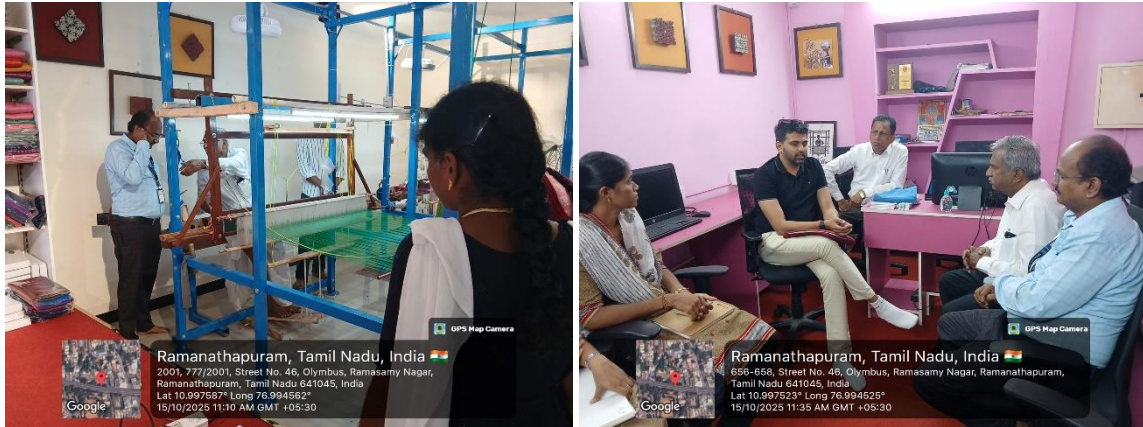
- Visited Dyer, weavers, warping artisan and cooperative society





Field visit to Silkville Handloom Institute:

- For exploring various methods and techniques used for weaver skill upgradation



Field visit to Jacquard Designer:

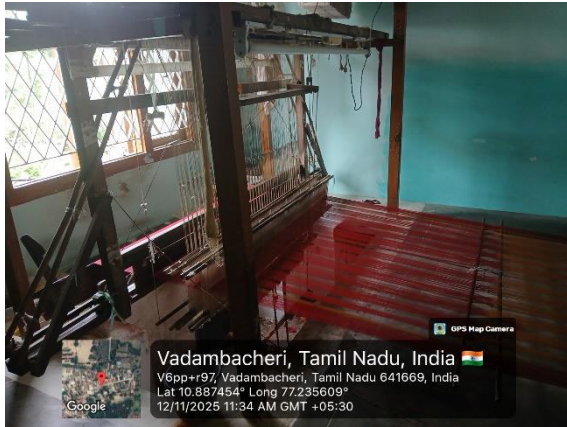
- Data collection regarding negamam jacquard designs

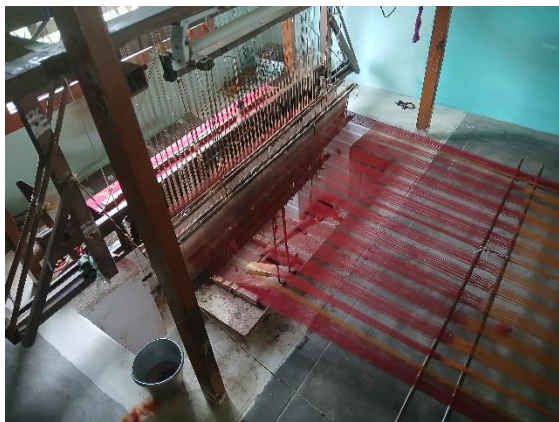


Vadambacheri Field visits:

- For Questionnaire Data collection



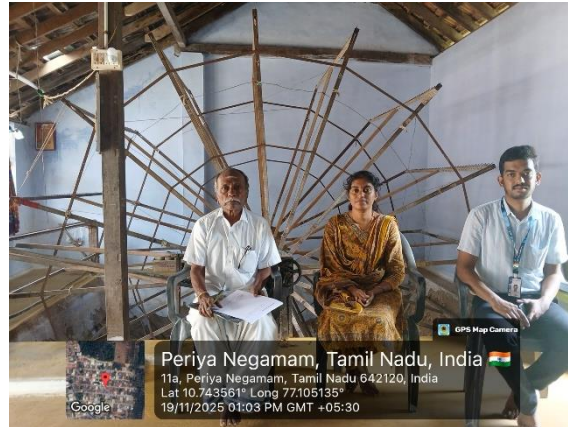






Negamam field visit:

- Data collection from warping artisan

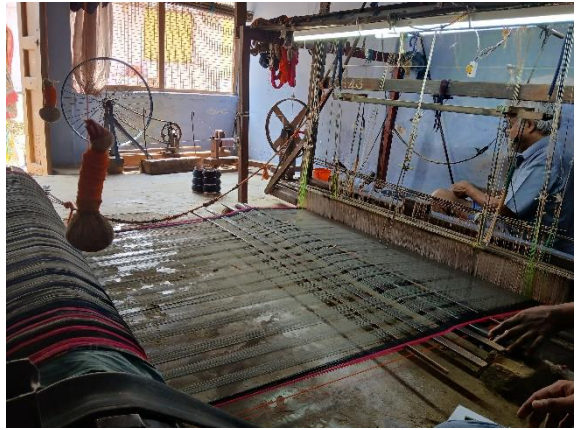
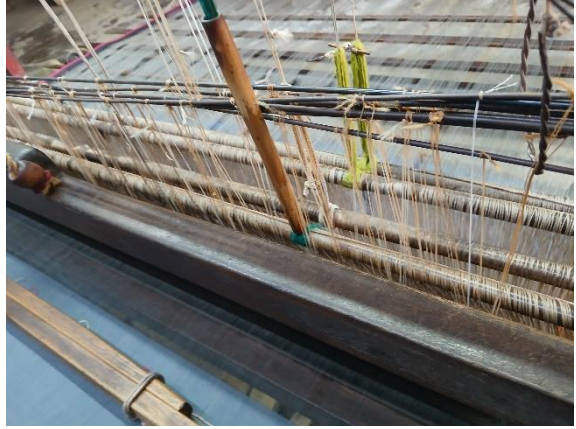


- Data collection from Master weavers



- Data collection from weavers







Varanasi field visit:



Field Visit photos of AI app development Photo Collection:



Reviving Negamam Heritage Sarees through Value Addition: A Capacity – Building Programme at Vadambacheri Society:





One day workshop on Pain-free weaving: Ergonomic training, Exercise and Health Camp (Negamam Clusters) at Kumaraguru college of technology:



Negamam Saree Photos:

The full product and close ups of the designs

Saree Design 1:



- Tower Komai



- Korvai



- Rudhraksha, Neli



- Normal Komai

Saree Design 2:



- Kattari and Button

Saree Design 3:

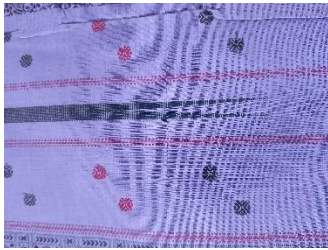
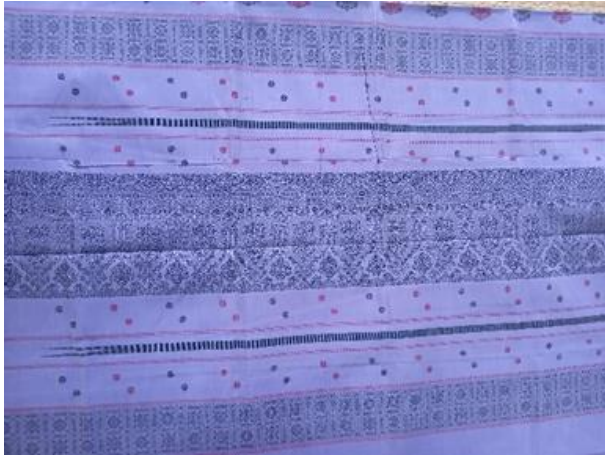


- Uruvam



- Moghu

Saree Design 4:



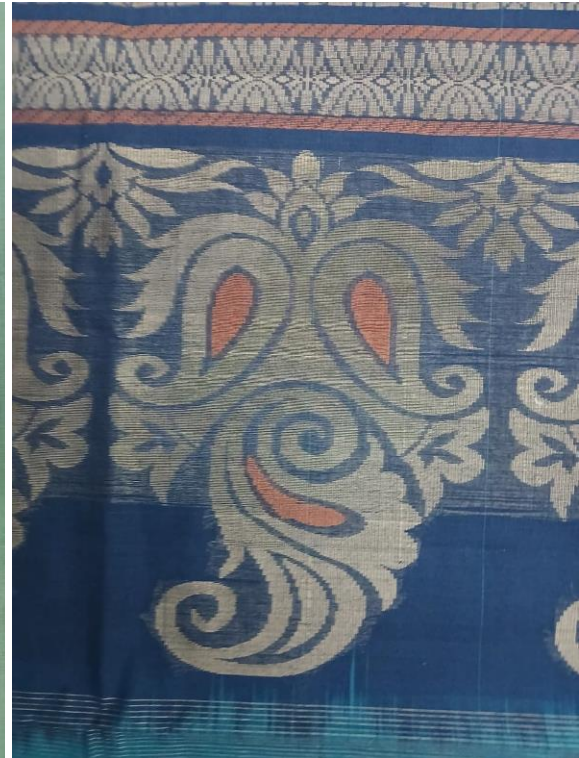
- Oosi korvai komai

Saree Design 5:





- Manga







Link More Photos: [Photos](#)

7. Videos (Short clips of weavers describing their journey)

Videos: https://kumaragurudtsteam-my.sharepoint.com/:f/g/personal/kavibharathi_m_txt_kct_ac_in/IgAarmWAsGFwSI2EKnupNM-EAVtC2cCEpt1oyTnSjdlcJkw?e=D8TLnq

8. Technological Intervention

Right now, there are not much technological interventions found in the negamam clusters and lots of scope for the following:

- **Sizing & Gumming**
 - Current: Traditional starch sizing and gumming
 - Gap: Inconsistent viscosity; yarn breakage
 - Direction: Comparative study of sizing and gumming methods
- **Dyeing Systems**
 - Current: Conventional dyeing methods

- Gap: Shade variation; compliance issues
 - Direction: Use of azo-free dye alternatives
- **Loom Ergonomics**
 - Current: Pit loom and raised setup
 - Gap: Weaver fatigue; posture strain
 - Direction: Ergonomic seating, treadle optimization, work-aid attachments
- **Dual-Axis Adjustable Warp Elevation Mechanism for Controlling Warp Geometry and Fabric Compactness in the Negamam Handloom Weaving**
 - Current: Fixed ~30 inches from cloth beam; ~2-inch lift
 - Gap: Limited comfort; poor pick control
 - Direction: Trial varied horizontal & vertical positions
- **Material Options**
 - Current: 80s cotton yarn
 - Gap: Limited sustainable options
 - Direction: Bamboo / regenerated cellulose Lyocell / Eri silk blends evaluation
- **Digital Printing**
 - Current: Jacquard Weaving
 - Gap: Limited exploration of alternative textile design techniques
 - Direction: Combine traditional Jacquard weaving with digital printing for innovative designs